

Mathematical Statistics Notes

Structured from Handwritten Notes

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1 Limiting Distribution

[1, pp. 1, 2]

1.1 Convergence Concepts

- **Limiting Distribution:** An important limit formula is $\lim_{n \rightarrow \infty} (1 + \frac{b}{n})^n = e^b$ [1, pp. 3, 5].
- **Convergence in Probability:** $Y_n \xrightarrow{p} y$, which implies $\lim_{n \rightarrow \infty} P(|Y_n - y| < \epsilon) = 1$ [1, pp. 7, 8]. This is also referred to as stochastic convergence ($Y_n \xrightarrow{p} c$) [1, p. 9].
- **Law of Large Numbers (LLN):** For sample mean $\bar{X}_n \xrightarrow{p} \mu$ given $E(X_i) = \mu$ [1, pp. 11, 12, 13].

Properties of Convergence

Given $X_n \xrightarrow{p} c$ and $Y_n \xrightarrow{p} d$ [1, p. 20]:

- $aX_n + bY_n \xrightarrow{p} ac + bd$ [1, p. 21]
- $X_n Y_n \xrightarrow{p} cd$ [1, p. 22]

Slutsky's Theorem: Given $X_n \xrightarrow{p} c$ and $Y_n \xrightarrow{d} Y$ [1, p. 27]:

- $X_n + Y_n \xrightarrow{d} c + Y$ [1, p. 28]
- $X_n Y_n \xrightarrow{d} cY$ [1, p. 29]
- $\frac{Y_n}{X_n} \xrightarrow{d} \frac{Y}{c}$ for $c \neq 0$ [1, p. 30]

1.2 Asymptotic Convergence & Central Limit Theorem

- **Asymptotic Convergence:** If $\frac{Y_n - \mu}{c/\sqrt{n}} \xrightarrow{d} Z \sim N(0, 1)$, then $Y_n \xrightarrow{p} \mu$ [1, pp. 16, 17]. Equivalently, for large n , Y_n is approximately distributed as $N(\mu, c^2/n)$ [1, pp. 31, 32, 34].
- **Central Limit Theorem (CLT):** $Z_n = \frac{\sum x_i - n\mu}{\sqrt{n}\sigma} = \sqrt{n} \frac{\bar{x}_n - \mu}{\sigma} \xrightarrow{d} N(0, 1)$ [1, p. 36].
- **Order Statistics:** The limiting distribution of central order statistics $X_{k:n}$ is asymptotically normal with mean x_p given $\frac{k}{n} \rightarrow p$ [1, pp. 35, 46].
- **Transformation Method:** For $Y = g(X)$, the PDF is $f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$ where $x = g^{-1}(y)$ [1, pp. 55, 56, 57, 62].

2 Point Estimation

[1, pp. 189, 190]

2.1 Methods of Estimation

- **Method of Moments Estimator (MME):** The j -th moment is $\mu'_j(\theta) = E[X^j]$ [1, pp. 191, 192]. We use the sample moments $M'_j = \frac{1}{n} \sum_{i=1}^n (X_i)^j$ to estimate [1, p. 194]. For example, $M'_1 = \bar{X}_n$ [1, p. 195].
- **Maximum Likelihood Estimator (MLE):** The likelihood function is $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$ [1, pp. 198, 199]. The MLE $\hat{\theta}$ satisfies $L(\hat{\theta}) = \max_{\theta \in \Omega} f(x_1 \dots x_n; \theta)$ [1, p. 200].
- **Invariance Property:** If $\hat{\theta}$ is the MLE of θ and $\tau(\theta)$ is a function of θ , then $\tau(\hat{\theta})$ is the MLE of $\tau(\theta)$ [1, pp. 197, 203].

2.2 Criteria for Estimators

Estimator Properties

- **Unbiased Estimator:** $E[T] = \tau(\theta)$ for all $\theta \in \Omega$ [1, p. 205].
- **UMVUE:** T^* is a Uniformly Minimum Variance Unbiased Estimator if for any other unbiased estimator T , $Var[T^*] \leq Var[T]$ [1, pp. 208, 209].
- **Cramer-Rao Lower Bound (CRLB):** The variance of an unbiased estimator T is bounded by $Var[T] \geq \frac{(\tau'(\theta))^2}{nE[(\frac{\partial}{\partial \theta} \log f(X; \theta))^2]}$ [1, pp. 211, 212].
- **Efficiency:** Relative efficiency is $re(T, T^*) = \frac{Var(T^*)}{Var(T)}$ [1, pp. 220, 221].
- **Mean Squared Error (MSE):** $MSE(T) = E(T - \tau(\theta))^2 = Var(T) + (b(T))^2$ where $b(T)$ is the bias [1, pp. 221, 222].

2.3 Large-Sample Properties

[1, p. 223]

- **Simple Consistency:** $\lim_{n \rightarrow \infty} P[|T_n - \tau(\theta)| \leq \epsilon] = 1$, denoted $T_n \xrightarrow{p} \tau(\theta)$ [1, p. 223].
- **MSE Consistency:** $\lim_{n \rightarrow \infty} E[T_n - \tau(\theta)]^2 = 0$ [1, p. 224]. If MSE consistent, it is also simply consistent [1, p. 229].
- **Asymptotically Unbiased:** $\lim_{n \rightarrow \infty} E[T_n] = \tau(\theta)$ [1, p. 225].

3 Sufficient & Complete Statistics

[1, p. 144]

3.1 Sufficient Statistics

- **Definition of Joint Sufficiency:** The conditional distribution of the sample given $S(\vec{x})$ is free of θ : $f(\vec{x}|\vec{s}) = \frac{f(\vec{x};\vec{\theta})}{f_s(\vec{s};\vec{\theta})}$ [1, pp. 145, 146].
- **Properties:** MLEs and Order Statistics are jointly sufficient [1, p. 148]. If $X_1 \dots X_n$ is a random sample, the order statistics form a jointly sufficient set for θ [1, p. 151].

Rao-Blackwell Theorem

Let T be an estimator with $E[T] = \tau(\theta)$. If S is sufficient, then $T^* = E[T|S]$ is a function of S , is an unbiased estimator of $\tau(\theta)$, and $Var(T^*) \leq Var(T)$ [1, pp. 152, 153].

3.2 Completeness & UMVUE

- **Completeness:** A family is complete if $E[u(T)] = 0$ for all $\theta \in \Omega \implies u(T) \equiv 0$ [1, pp. 154, 156].
- **Lehmann-Scheffé Theorem:** If S is a vector of jointly complete and sufficient statistics, and $T^*(S)$ is an unbiased estimator for $\tau(\theta)$, then T^* is the unique UMVUE [1, pp. 159, 161, 162].

Regular Exponential Class (REC) & Basu's Theorem

REC: $f(x; \theta) = c(\theta)h(x) \exp[\sum_{j=1}^k q_j(\theta)p_j(x)]$ [1, p. 164].

Basu's Theorem: If S is a complete and sufficient statistic for θ , and T is an ancillary statistic whose distribution does not depend on θ , then T and S are stochastically independent [1, pp. 166, 168, 170].

Example (Normal Distribution): For a sample from $N(\mu, \sigma^2)$, the joint PDF can be factored into $g(s_1, s_2; \mu, \sigma^2)h(x_1 \dots x_n)$, showing $s_1 = \sum x_i$ and $s_2 = \sum x_i^2$ are jointly sufficient [1, pp. 176, 177, 181, 182].

4 Confidence Interval & Hypothesis Testing

[1, p. 70]

4.1 Confidence Interval (CI)

- A $100\gamma\%$ confidence interval for θ is $(L(X_1 \dots X_n), U(X_1 \dots X_n))$ such that $P[L(X_1 \dots X_n) \leq \theta \leq U(X_1 \dots X_n)] = \gamma$ [1, pp. 73, 74].
- **Pivotal Quantity:** $Q = q(X_1 \dots X_n; \theta)$ is a pivotal quantity if its PDF does not depend on θ or any other unknown parameters [1, pp. 79, 81].
- **Examples:**
 - For $N(\mu, \sigma^2)$, $\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \sim t(n-1)$ [1, p. 84].
 - For $EXP(\theta)$, $\frac{2n\bar{X}_n}{\theta} \sim \chi^2(2n)$ [1, p. 85].

4.2 Hypothesis Testing

[1, p. 119]

- **Errors:**
 - Type I Error (α , significance level): $\alpha = P[\text{Reject } H_0 | H_0 \text{ is true}] = P[|\bar{X} - \mu_0| > c | \mu = \mu_0]$ [1, pp. 122, 125].
 - Type II Error (β): $\beta = P[\text{Accept } H_0 | H_1 \text{ is true}]$ [1, p. 126].

Most Powerful Test & Neyman-Pearson

Neyman-Pearson Lemma: For testing simple hypotheses, the most powerful test rejects H_0 in favor of H_1 if the likelihood ratio $\lambda = \frac{f(x; \theta_0)}{f(x; \theta_1)} \leq k$ [1, pp. 133, 137].

- **Likelihood Ratio Test (LRT):** For composite hypotheses, the generalized test statistic is $\lambda = \frac{\max_{\theta \in \Omega_0} f(\vec{x}; \theta)}{\max_{\theta \in \Omega} f(\vec{x}; \theta)}$ [1, p. 142].

References

- [1] Rachel Lei, *PTS3 Handwritten Course Notes*, unpublished course notes, University of Amsterdam, n.d.